

FINAL EXAM—FALL 2025

ECO403/503—INTRODUCTION TO CONTRACT THEORY AND AUCTIONS

DECEMBER 5, 2025

Final Exam Instructions and Rules

Please, read this carefully before attempting anything!

The final examination consists of two parts:

- A **take-home list of problems**, to be completed in groups.
- A **group oral examination**, during which *each student will be asked individual questions* to assess their personal understanding and contribution.

Collaboration and Submission

- Students must work in **groups of up to 5 members**. A deviation of +1 is allowed *only with prior written consent* from the instructor.
- Only **one written report** is required per group.
- Each group must send **one email** to the instructor stating the **full list of group members**.
- **Inter-group consultation is permitted**, but **each group must write their own solutions independently**.
- Students are **strictly prohibited** from discussing this exam with **anyone outside the class**, or from sharing oral exam content with others after their interview.
- **All written submissions must be typed**, including all mathematical expressions.
 - It is strongly recommended to use $\text{\LaTeX}$ (e.g., Overleaf) or Microsoft Word with an equation editor.
 - **Handwritten mathematics embedded as images will not be accepted.**
 - This is a formal report and should be treated accordingly.
 - The submission must consist of a single PDF file uploaded to Moodle.
 - Your submission must be received **before 12:00 (noon) on Wednesday, December 8, 2025**. Only one submission per team.
 - Your written solutions should be **careful, rigorous, and succinct**. There are **penalties for unnecessarily long or poorly structured arguments**. Knowing how to express a mathematical argument clearly and concisely is part of what is being evaluated.

Use of Resources

- You may consult:
 - Your own class notes and printouts of the slides.
 - Any book listed as a reference in the syllabus.
 - Any other book or online source you deem appropriate.
- **All references must be properly cited and acknowledged.**
- You may use **any computer algebra system** (e.g., Mathematica, WolframAlpha, Maple) for final calculations. However, you must provide a **complete logical argument**, omitting only routine algebra when appropriate.

Use of Generative AI Tools

- Use of **Generative AI tools** (e.g., ChatGPT, Claude, Copilot, DeepSeek, etc.) is permitted **only up to 25%** of your written submission.
- You are fully responsible for the **correctness and understanding** of any AI-generated content.
- During the oral exam, I will assess whether each student understands the material. If it becomes clear that AI did the thinking, **this constitutes academic dishonesty**.

Integrity and Consequences

- Uploading the exam paper to any internet repository or database is strictly prohibited.
- Any violation of these rules will be penalized under the Academic Integrity Policy. Even a **Category A violation** may result in **failing the course**, at my discretion and that of the Academic Discipline Committee.

Instructions and Grading Criteria

The exam contains **200 total points**, but the final score will be computed **over 100**. Exceptional performance therefore yields extra credit towards your final exam grade.

It is **REQUIRED** to attempt **at least one problem from each section**. Failure to do so results in an automatic significant penalty.

Section 1 (Vignettes)

- You must choose **exactly one vignette** from Section 1 to prepare for the oral exam.
- **No written submission is required.**
- The chosen vignette is worth **20 points**, evaluated entirely during the oral exam.
- This is the *entry fee* to the exam: every group must be ready to discuss one vignette thoughtfully.

Section 2 (Knowledge)

Problem 2.1 = 30 points

Problem 2.2 = 20 points

Section 3 (Bayesian Games and Beliefs)

Problem 3.1 = 30 points

Problem 3.2 = 30 points

Section 4 (Auctions)

Problem 4.1 = 30 points

Problem 4.2 = 40 points

Requirement to Attempt Every Section

To discourage strategic skipping:

- You must submit a **complete solution to at least one problem from Section 2, one from Section 3, and one from Section 4.**
- You may attempt additional problems for more points.
- Only **fully attempted** problems will be graded.

Anti-Point-Farming Policy

- **Partial attempts are penalized.** If you choose to attempt a problem, you must attempt it *in full*. Submitting fragments, isolated equations, undeveloped ideas, or lists of steps without complete arguments will be penalized more heavily than leaving the problem blank.
- **Complete solutions** are graded for **correctness, clarity, and rigor.**
- **Incomplete solutions**—including missing steps, unjustified claims, skipped cases, or logically unfinished arguments—receive large deductions.
- **Do not attempt problems you do not understand simply to accumulate points.** Low-quality or AI-generated work that you cannot explain during the oral exam will damage your grade severely and may trigger an academic integrity investigation.

- **LLM misuse is self-destructive.** If you cannot explain your own written solution during the oral exam, the penalty is severe.

Final Notes

Students earning exceptional scores (significantly above 100/100) will receive:

- My **personal distinction** for outstanding performance,
- Eligibility for **Teaching Assistant** positions,
- Priority for **Research Assistant** positions,
- Priority for **MA supervision**,
- And the instructor's long-term admiration.

Acknowledgements

Several problems in this examination were developed with the valuable assistance of the Teaching Assistants for this course. Their insight, creativity, and dedication to academic rigor contributed meaningfully to the design and refinement of the problems. I am grateful for their thoughtful suggestions and for the care with which they approached the task.

Problems in the next page. Good luck!



1 Auction Vignettes: Overall Evaluation of Course Content

This task contains three real-world-inspired vignettes about auctions in Kazakhstan. Your task is to read and reflect deeply on each situation. During the oral examination, I will ask questions derived from these vignettes. You may be asked to:

- Discuss possible auction models and information structures.
- Present informal or formal arguments.
- Evaluate strategic behavior of agents.
- Identify connections to main course topics and the concepts we covered within, such as **Knowledge & Beliefs**, **Bayesian Games**, **SIPV Auctions**, **Revenue Equivalence**, **Reserve Prices & Entry Fees**, and **Interdependent Value Auctions**.

Think of this as presenting a *business case*—because I know just how much you adore those. Be ready to discuss each vignette at three levels: **informally** (as if you’re briefing a non-economist director who just wants “the takeaway”), **semi-formally** (as if advising a competent economist who actually understands the mechanics), and a bit more **formally**—say, if you were trying to impress a sharp professor...like me.

You are not required to submit anything written for this task. But be ready to think precisely, speak clearly, and—if asked—write on the whiteboard.

1.1 The Privatization Puzzle—Kazakhstan, 1994

In September 1994, Kazakhstan auctioned off over 1,600 state-owned enterprises as part of its rapid privatization program. The format was an **open outcry ascending auction**. Prices rose dramatically — in some cases, sevenfold. The result caused widespread concern: some bidders claimed they had no reliable data on firms and regretted their bids. Subsequently, the government reverted to **sealed-bid auctions**, citing instability and overvaluation.

1.2 Refining the Bid—The Stratus Oil Auction (2022–2024)

In 2022, Mangystau Oil Refining (MOR) acquired the bankrupt Stratus Oil refinery in Turkestan through a **public auction**. Many bidders exited early, citing a lack of transparency and concerns over environmental liabilities. MOR submitted a strong bid and was declared the winner. However, in 2024, the Kazakh court invalidated the deal, stating the auction had failed to produce a competitive or legitimate outcome and lacked procedural safeguards.

1.3 Wind Power for Sale—Renewable Energy Auctions (2018–2022)

To attract investment in green energy, Kazakhstan held a series of **reverse auctions** for wind and solar power. Bidders submitted sealed bids indicating the price at which they were willing to sell electricity to the grid. Winning bids were often 30–40% below previous tariffs. Some winning firms later struggled to deliver on their commitments, raising questions about “overly aggressive bidding” and whether auction incentives had induced inefficiencies or unsustainable outcomes.

2 Knowledge

2.1 Fun with the standard model of Knowledge

Consider¹ a knowledge space $(N, \Omega, (\mathcal{P}_i)_{i \in N})$, where N and Ω are finite sets. Let K_i be the knowledge operator that we studied in class, i.e., for an event $A \in 2^\Omega$, $K_i(A) = \{\omega \in \Omega : P_i(\omega) \subseteq A\}$.

- 2.1.1. Give an example of a knowledge space and an event A for which a player does not know A nor A^c at any state of the world.
- 2.1.2. Derive the necessary and sufficient conditions for an event A for which a player does not know A nor A^c at any state of the world.
- 2.1.3. Derive the necessary and sufficient conditions for information partitions $\mathcal{P}_i$ to satisfy the property that there always exist an event A such that player i does not know A nor A^c in any of the states of the world.

¹This problem was developed with the assistance of **Abuzer Abuov**, whose taste in mathematical problems may serve as inspiration.

2.2 Sherlock Holmes and the Even More Curious Case of the Dog and the Cat

Sherlock Holmes once investigated a disturbance at a rural state in which he noted an even more curious incident, that of the dog that barked and the cat that howled in the night. When Watson objected that the dog did not bark and the cat did not howl, Holmes replied “that is the curious incident to which I refer.” Reviewing the events of the night, Holmes asked Watson to observe that neither animal had made any sound. Watson considered this unremarkable, but Holmes pointed out that the silence was the decisive fact: a human intruder would necessarily have triggered the dog, and a canine intruder would necessarily have provoked the cat. Therefore, the joint absence of signals ruled out both forms of intrusion, leaving only the possibility that no intruder had entered.

We now formalize this situation from the perspective of an observer who, like Watson, correctly interprets positive signals but does not draw any inference from silence. Let $\Omega = \{a, b, c\}$, where a denotes the presence of a human intruder, b the presence of a dog intruder, and c the absence of any intruder. The animals behave deterministically: the dog signals in state a and only in state a ; the cat signals in state b and only in state b ; in state c both remain silent.

The observer identifies states a and b perfectly whenever the corresponding signal occurs, but treats silence as compatible with all three states. The possibility correspondence is therefore $P(a) = \{a\}$, $P(b) = \{b\}$, $P(c) = \{a, b, c\}$. The associated knowledge operator is $K(E) = \{\omega \in \Omega : P(\omega) \subseteq E\}$, and complements are taken relative to Ω .

- 2.2.1. Compute the values of $K(E)$ for all nontrivial events: $\{a\}$, $\{b\}$, $\{c\}$, $\{a, b\}$, $\{a, c\}$, $\{b, c\}$, and interpret the observer’s knowledge in each case. Identify at which states the observer (Watson) knows that a human intruder entered, knows that a dog intruder entered, knows that no intruder entered, and knows that a particular type of intruder did not enter.
- 2.2.2. Show that negative introspection fails for the event $\{a\}$. Identify a state ω such that $\omega \in K(\{a\}^c)$ (the observer does not know a), yet $\omega \notin K(K(\{a\}^c))$ (the observer does not know that he does not know a). Provide a brief explanation of why Watson’s interpretation of silence generates this failure.
- 2.2.3. Define the lack-of-knowledge operator

$$L(E) = K(E)^c, \quad L^n(E) = L(L^{n-1}(E)) \quad (n \geq 2).$$

For $E = \{a\}$, compute

$$L(\{a\}), \quad L^2(\{a\}), \quad L^3(\{a\}), \quad L^4(\{a\}),$$

describe any pattern that emerges, and provide concise interpretations of $L(\{a\})$, $L^2(\{a\})$, and $L^3(\{a\})$ in terms of the observer’s increasingly higher-order lack of knowledge about whether a human intruder entered. Identify any state that lies in every set $L^n(\{a\})$.

- 2.2.4. Define for each event $E \subseteq \Omega$ the *unawareness* operator

$$U(E) = \bigcap_{n=1}^{\infty} L^n(E).$$

Compute $U(\{a\})$, as well as $U(\emptyset)$, $U(\Omega)$, $U(\{c\})$, and $U(\{a, b\})$. Then determine $U(\{b\})$, $U(\{a, c\})$, and $U(\{b, c\})$. Summarize your results in a table listing $U(E)$ for every $E \subseteq \Omega$.

- 2.2.5. Provide a precise interpretation of the condition $\omega \in U(E)$ and explain why it might be viewed as a state of persistent lack of knowledge about E . Identify all events for which $U(E) \neq \emptyset$ and relate your answer to the silent state in Holmes and Watson story from above. For $E = \{a\}$, compute $U(U(E))$ and compare it with $U(E)$; discuss whether it is conceptually coherent for an observer to be “unaware of E ” without being “unaware of being unaware of E .” Conclude with a brief assessment of whether U provides a coherent representation of unawareness in this information structure.

3 Beliefs & Bayesian Games

3.1 The Monty Hall Problem as a Bayesian Game

The Monty Hall problem comes from the television show *Let’s Make a Deal*. A contestant chooses one of three doors; one hides a car and the other two hide goats. After the initial choice, the host, Monty Hall, typically opens one of the *other* doors to reveal a goat and offers the contestant the option to *switch*. The standard analysis, popularized by Marilyn vos Savant², shows that switching yields a probability 2/3 of winning the car, while staying

²See Marilyn vos Savant (1990), “Ask Marilyn,” *Parade Magazine*, September 9, 1990.

yields only $1/3$. In contrast, the folk solution incorrectly claims that once one door is opened the problem becomes a $1/2$ versus $1/2$ choice, implying indifference between switching and staying.

In practice, Monty Hall did not always behave mechanically. His choices on the actual show occasionally deviated from the textbook assumptions: sometimes he declined to open a door; sometimes he appeared to act strategically to increase suspense. To capture this feature, you will study a Bayesian game in which Monty may be either *sympathetic* or *antipathetic*, and where his actions may or may not reveal information about whether the contestant's initial choice was correct.

There are two players, Amy (the player), and Monty Hall (the host).

Nature first determines whether Amy's initial choice is correct:

$$\theta_1 = \text{"Amy picked the car"}, \quad \theta_2 = \text{"Amy picked a goat"}.$$

Nature chooses θ_1 with probability $1/3$ and θ_2 with probability $2/3$. Monty observes θ ; Amy does not.

Monty's motive may be sympathetic (M_S) or antipathetic (M_A), with probabilities

$$p = \Pr(M = M_S), \quad 1 - p = \Pr(M = M_A),$$

common knowledge. A state is the pair (θ, M) .

After observing (θ, M) , Monty chooses either to reveal a goat (r) or to reveal nothing (h). If r is chosen, Amy chooses whether to switch (s) or keep (k). If h is chosen, she must keep.

Amy observes only Monty's action. Her strategy specifies what she does after r ; if h occurs, she keeps automatically.

Amy's payoff is 1 if she wins the car and -1 otherwise. Monty's payoff depends on whether he is sympathetic or antipathetic. Payoffs (Amy listed first) follow.

Sympathetic Monty M_S :

		r	h			r	h
θ_1 :	s	$-1, 0$	$1, 1$	θ_2 :	s	$1, 2$	$-1, -1$
	k	$1, 2$	$1, 1$		k	$-1, 0$	$-1, -1$

Antipathetic Monty M_A :

		r	h			r	h
θ_1 :	s	$-1, 2$	$1, -1$	θ_2 :	s	$1, 0$	$-1, 1$
	k	$1, 0$	$1, -1$		k	$-1, 2$	$-1, 1$

3.1.1. Define the state space and write the prior distribution over states. Describe what each player observes at each point in the game.

3.1.2. (a) Describe formally the set of pure strategies for Monty.

(b) Describe formally the set of pure strategies for Amy.

3.1.3. (**Sympathetic Monty, $p = 1$**)

(a) Compute a Bayesian Nash equilibrium. For each $\theta \in \{\theta_1, \theta_2\}$, determine whether Monty chooses r or h , and what Amy does after observing r .

(b) Does Monty's action reveal information about whether Amy's initial choice was correct? Explain.

3.1.4. (**Antipathetic Monty, $p = 0$**)

(a) Show that no Bayesian Nash equilibrium in pure strategies exists.

(b) Find a mixed-strategy Bayesian Nash equilibrium in which Monty's type (θ_2, M_A) mixes between r and h , and Amy mixes between s and k after observing r . Derive the mixing probabilities.

(c) In the mixed equilibrium, compute the posterior probability $\Pr(\theta_1 | r)$ and explain why Amy is indifferent between switching and keeping.

3.1.5. (**General motive uncertainty, $p \in (0, 1)$**)

(a) Identify conditions on p under which a Bayesian Nash equilibrium exists in which Amy switches with probability 1 after observing r .

(b) Describe how the equilibrium changes when p is sufficiently small. Discuss whether equilibrium beliefs and actions can reproduce the folk " $1/2$ - $1/2$ " intuition.

3.1.6. Explain how this game-theoretic model helps reconcile the classical "always switch" result with the widespread intuitive error that the problem is "50-50." Comment on Monty's strategic behavior, Amy's uncertainty about his motive, and the informational content of r .

3.2 Megumi's Overkill Problem and Bayesian Updating

Megumi³ faces a sequence of independent duels against curses. Before each duel he must choose one of his shadow summons. Each summon has an integer level $s \in \{1, \dots, S\}$, where higher levels are stronger but impose a recoil cost $c(s)$ with $c'(s) > 0$. Level S is catastrophic: even if Megumi wins he dies, so $c(S) = D > 0$. Megumi never observes the opponent's strength ex ante. For each level s , let

$$p_s = \Pr(\text{enemy is weak enough to be defeated by level } s).$$

His only signal is a Bernoulli outcome: a win (signal 1) occurs with probability p_s and a loss (signal 0) occurs with probability $1 - p_s$.

For each summon level he begins with a prior $p_s \sim \text{Beta}(\alpha_s, \beta_s)$. After using level s in n past duels and observing w wins and ℓ losses, conjugacy of the Beta prior with Bernoulli signals implies

$$p_s \mid (w, \ell) \sim \text{Beta}(\alpha_s + w, \beta_s + \ell), \quad \mu_s = \mathbb{E}[p_s \mid w, \ell] = \frac{\alpha_s + w}{\alpha_s + \beta_s + w + \ell}. \quad (1)$$

If Megumi loses he dies and receives payoff $-D$. If he wins using $s < S$, he suffers recoil cost $c(s)$. If he wins using $s = S$, he dies. Hence $U(S) = -D$, and for $s < S$,

$$U(s) = \mu_s(-c(s)) + (1 - \mu_s)(-D).$$

- 3.2.1. The posterior mean above shows how Bernoulli evidence affects Megumi's beliefs. An additional win increases μ_s , while an additional loss decreases it. Intuitively, repeated survival using low levels pushes Megumi toward the view that most opponents are weak, reducing his perceived need for high-level summons. Please derive the expressions in equation 1.
- 3.2.2. Derive the expected utility difference $U(s) - U(s + 1)$ for $s < S - 1$. Characterize when Megumi strictly prefers the lower level.
- 3.2.3. Show that for any posterior $\mu_S < 1$, the level S is strictly dominated by some level $s < S$. Provide an interpretation.
- 3.2.4. Now suppose Megumi faces a particular curse that has already defeated k skilled sorcerers, each of whom used the same summon level s . Model each past duel as a Bernoulli trial: conditional on p_s , the curse survives a duel (the sorcerer loses) with probability $1 - p_s$, independently across duels. Assume Megumi's prior for p_s is $\text{Beta}(\alpha_s, \beta_s)$.

- (a) Let $\mathcal{S}_k$ denote the event that the curse survives k duels in a row. Write the likelihood $\Pr(\mathcal{S}_k \mid p_s)$ and use Bayes' rule to derive the posterior distribution of p_s given $\mathcal{S}_k$. Show that

$$p_s \mid \mathcal{S}_k \sim \text{Beta}(\alpha_s, \beta_s + k).$$

- (b) Define

$$q_s(k) = \Pr(\text{enemy is weak vs } s \mid \mathcal{S}_k) = \mathbb{E}[p_s \mid \mathcal{S}_k].$$

Compute $q_s(k)$ explicitly and compare it to the prior mean

$$\mu_s^{\text{prior}} = \mathbb{E}[p_s] = \frac{\alpha_s}{\alpha_s + \beta_s}.$$

Show that $q_s(k) < \mu_s^{\text{prior}}$ for all $k \geq 1$.

- (c) Define the expected overkill at level s as

$$\text{Overkill}(s) = c(s) [\mu_s^{\text{prior}} - q_s(k)].$$

Interpret this expression as recoil damage generated by relying on the unconditional belief μ_s^{prior} when the relevant conditional belief $q_s(k)$, given the curse's survival record, is lower. Explain why this is analogous to a winner's-curse effect.

- 3.2.5. Consider the following comparative statics.

- (a) Show that as w becomes large relative to ℓ , the posterior mean μ_s increases. Determine how Megumi's optimal summon level changes as a result.
- (b) Show that as k increases, the conditional belief $q_s(k)$ decreases. Determine how Megumi's optimal summon level responds to increases in k .

³This problem was developed with the assistance of **Sultan Yessirkep**, whose erudition in anime and remarkable creativity are inspiring.

4 Auctions

4.1 Common Value First-Price Auction: Exploring the Winner's Curse and Regret

Consider a sealed-bid *first-price auction* with two bidders, where the *common value* of the object is given by $V = X_1 + X_2$, with X_1 and X_2 being independent and identically distributed random variables on the interval $[0, 1]$, with cumulative distribution function F that is commonly known among the bidders.

Each bidder $i \in \{1, 2\}$ privately observes their own signal X_i , and knows the distribution of the other bidder's signal.

Assume the distribution function is $F(x) = x^\theta$, for $x \in [0, 1]$, where $\theta > 0$, with corresponding density $f(x) = \theta x^{\theta-1}$.

Each bidder submits a bid based on their private signal. The highest bidder wins the object and pays their own bid.

- 4.1.1. Compute the symmetric, monotonically increasing, and differentiable equilibrium bidding function of the first-price auction. Analyze the bidding function with respect to the parameter θ . Are there any cases in which a bidder bids above her own signal?
- 4.1.2. Compute $\mathbb{E}[V \mid X_1 = x, Y_1 < x]$, where Y_1 is the highest order statistic from the sample excluding player 1. Use this to argue that, no matter the value of θ , player 1 is never a victim of the winner's curse at the interim stage when using the symmetric equilibrium strategy.
- 4.1.3. Suppose that both players follow the equilibrium strategies, and that after the auction concludes, player 1 is the winner and the bid of player 2 is also announced. For each value of θ , characterize the region of regret for player 1: the pairs (x, y) such that player 1 wins but regrets doing so because they overpaid. Compute both the interim probability of regret and the ex-ante probability of regret for player 1. Does this contradict the assertion of the previous item?

4.2 Course Registration at NU as a Multi-Unit All-Pay Auction

Registration⁴ for popular courses at Nazarbayev University is often perceived by students as stressful and unpredictable. In practice, the registration system has many complicating features: students register for multiple courses simultaneously; schedules create complementarities and conflicts; some students have priority due to major or seniority; and differences in internet speed or access to faster computers influence how quickly one can submit a registration request. In reality, students are bidding for packages of courses with important complementarities across time slots and degree requirements.

In this problem you will consider a simplified abstraction. You will focus on a single popular course with capacity K seats, where $1 \leq K < n$. Each of the n interested students desires at most one seat. All scheduling, prerequisite, and multi-course interactions are ignored. Students choose costly effort in order to obtain one of the available seats. Effort corresponds to time spent refreshing the system, lost sleep, stress, or paying for a faster internet connection, and is paid regardless of success.

We model this as a K -prize all-pay auction: each student chooses an effort level, the highest K efforts receive seats, ties are broken uniformly at random, and all students pay their effort cost.

Each student i receives a signal $X_i \in [0, \bar{\omega}]$. Let $X = (X_1, \dots, X_n)$ and X_{-i} denote the signals of the other students. The value of a seat for student i is

$$V_i = u(X_i, X_{-i}),$$

where u is continuous, non-negative, strictly increasing in each argument, symmetric in the last $n - 1$ arguments, and satisfies $\mathbb{E}[V_i] < \infty$. The joint density $f(X_1, \dots, X_n)$ is symmetric and affiliated. Students are risk neutral and have quasi-linear utility. Each chooses an effort $e_i \geq 0$. Effort cost is given by a common, strictly increasing, strictly convex, twice continuously differentiable function $c : [0, \infty) \rightarrow [0, \infty)$ with $c(0) = 0$.

Seats are awarded to the K highest efforts. Ties are broken uniformly at random. Everyone pays $c(e_i)$ independently of the outcome.

Fix student 1. Let $X_{-1} = (X_2, \dots, X_n)$, ordered in decreasing order as $Y_1 \geq Y_2 \geq \dots \geq Y_{n-1}$.

Define $Z := Y_K$ K -th highest signal among the other students. In any symmetric strictly increasing equilibrium, student 1 with signal x obtains a seat exactly when $x \geq Z$.

Let $g_K(z \mid x)$ denote the conditional density of Z given $X_1 = x$, and $G_K(z \mid x)$ the associated distribution function,

$$G_K(z \mid x) = \mathbb{P}(Z \leq z \mid X_1 = x).$$

⁴This problem was developed with the assistance of Akmaral Mussayeva, whose eye for discovering economic problems in the wild is truly inspiring.

Let the conditional hazard rate be

$$\tau_K(z | x) = \frac{g_K(z | x)}{1 - G_K(z | x)}.$$

Define the conditional expected value

$$v(x, z) = \mathbb{E}[V_1 | X_1 = x, Z = z].$$

The following properties may be used without proof.

Fact 1 The pair (X_1, Z) is affiliated.

Fact 2 For each fixed z , the function $x \mapsto G_K(z | x)$ is non-increasing.

Fact 3 For each fixed z , the function $x \mapsto \tau(z | x)$ is non-increasing.

Fact 4 For each fixed x , the function $z \mapsto \tau(z | x)$ is non-decreasing.

Fact 5 The conditional value $v(x, z)$ is strictly increasing in both arguments.

A student who obtains a seat and exerted effort e_i receives payoff $V_i - c(e_i)$. A student who does not obtain a seat receives payoff $-c(e_i)$. We focus on symmetric, strictly increasing, differentiable effort strategies $\alpha(\cdot)$.

4.2.1. Assume $V_i = X_i$ and that $X_1, \dots, X_n$ are i.i.d. with distribution function F and density f . Effort cost and the allocation rule are as above.

- (a) For a symmetric strictly increasing effort strategy $\alpha(\cdot)$, express the probability that a student with value v obtains a seat in terms of F , n , and K .
- (b) Write the expected payoff of a student with value v who chooses effort e , assuming all others follow $\alpha(\cdot)$.
- (c) Impose optimality of choosing $e = \alpha(v)$ and derive the first-order condition characterizing $\alpha(v)$.

4.2.2. Return to the full interdependent-values environment and suppose students use a symmetric strictly increasing differentiable strategy $\alpha(\cdot)$.

- (a) Using the notation Z , $G_K(\cdot | x)$, and $g_K(\cdot | x)$, express the probability that a student with signal x obtains a seat when using effort $\alpha(x)$.
- (b) Express the expected payoff of a student with signal x who exerts effort $\alpha(x)$ in terms of $v(x, z)$, $g_K(z | x)$, and $G_K(z | x)$.
- (c) Derive the first-order condition characterizing the symmetric equilibrium effort function $\alpha(x)$. Indicate where the Facts are used to justify differentiability, monotonicity, and the validity of the conditioning steps.

4.2.3. Provide an interpretation in the context of NU course registration.

- (a) Explain why, in the interdependent-values environment, the optimal effort of a student with signal x differs from the naive best response that treats the value of a seat as independent of the event of being among the top K efforts. In particular, describe how the condition $x \geq Z$ (with Z the K -th highest competing signal) affects the student's expected value of obtaining the seat.
- (b) Discuss how equilibrium effort varies with the number of seats K . Explain why increasing K weakens the informational content of the event $x \geq Z$ and therefore reduces effort, while decreasing K strengthens this informational effect and leads to higher effort.